

JEE MAIN REVISION MASTER: UNITS & MEASUREMENTS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 227 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Units And Measurements - Core Application	227 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Units And Measurements - Core Application (Very High Yield — 227 PYQs)

Core Principle: Key problem pattern in Units And Measurements.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Ea..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "The density ρ of a uniform cylinder is determined by measuring its mass m , length l and diameter d . The measured values of $m, l, \dots$ " [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7° ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance?

- (A) 200α
- (B) 400α **[Correct]**
- (C) 300α
- (D) 500α

Correct Answer: B

Solution Analysis: To find the distance between two points when an object moves at a constant speed, we use formula, Distance = Speed $\times$ Time α is a new unit of length such that 1 unit (α) = speed of light in vacuum (c). The speed of light c is the distance light travels per unit of time (usually meters per second). $1\alpha = c \times (1\text{second})$ The time taken for light to travel from Venus to Earth is 6 minutes and 40 seconds. 1 minute = 60 seconds 6 minutes = $6 \times 60 = 360$ seconds Total time $t = 360\text{s} + 40\text{s} = 400$ seconds Using distance formula: Distance = Speed of light $\times$ Time Distance $= c \times t$